Abbas Taqvi

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Professional Summary

- Software engineer with a data infrastructure background, pursuing an M.S. in Data Science at Milano-Bicocca. Built a Rust ETL framework that reduced processing time from 500 to 50 minutes, backend services exposed through REST APIs, and production Kubernetes infrastructure. Seeking a software engineering internship in Italy.

Education

- **Università degli Studi di Milano-Bicocca** — Milan, Italy
  Master of Science in Data Science — *Oct 2025 – Present*
- **Habib University** — Karachi, Pakistan
  Bachelor of Science in Computer Science; GPA: 3.31/4.0; 100% merit scholarship (HU-TOPS) — *Aug 2020 – May 2024*

Experience

- **Logiciel Services LLC** — United States (Remote)
  Data Engineer I — *May 2024 – Nov 2025*
  - **Ferroflow: Rust ETL Framework**: Designed reusable processors and composable pipelines with chunking, streaming and memory-mapped file processing. Replaced a legacy .NET utility, reducing daily FIX-log processing from 500 to 50 minutes (10× faster).
  - **Backend Data Platform**: Built Kafka ingestion, Spark (Scala) processing, S3 storage, Trino queries and Airflow orchestration; simplified a 12-component architecture to 5. Added schema control, file compaction and partitioning for reliability.
  - **Text2SQL API**: Built a Python multi-agent chatbot with LangGraph and Trino, exposing natural-language queries through a REST API and reducing analyst turnaround by 41%.
- **Pullflow Inc.** — San Jose, CA (Remote)
  DevOps Intern — *Jul 2023 – Aug 2023*
  - **Kubernetes Migration**: Migrated production services from AWS EKS to on-premise Kubernetes (MicroK8s): wrote Dockerfiles, Helm charts and a Traefik reverse-proxy config, and deployed a self-hosted analytics dashboard with a zero-downtime cutover.
- **United Bank Limited** — Karachi, Pakistan
  Data Engineering Intern — *May 2023 – Jun 2023*
  - **Migration Pipeline**: Built a cross-region migration pipeline in Informatica and validated ingestion across 3 countries.
- **ImproData** — Pinedale, WY (Remote)
  Junior Data Engineer — *Jan 2021 – May 2022*
  - **Data Pipelines**: Built pipelines on Azure Databricks (PySpark) and automated ingestion via GCP Cloud Functions (JSON → GCS → BigQuery), eliminating about 3 hours of daily manual work.

Projects

- **Yohsin: Zero-Copy Serializer (Rust)**: Built a multi-threaded serializer/deserializer in 24 hours, sustaining about 1M records/s (2.25 GB/s). Used memory-mapped files and async I/O for O(1) random access; added write-ahead logging with recovery under 50 ms and benchmark suites.
- **Bayaaz: Published Mobile App**: Published a Flutter poetry app on Google Play and the App Store; maintain a stateless FastAPI backend with PostgreSQL, exposed through Cloudflare tunnels, and a React frontend on Cloudflare Pages.

Skills

- **Languages**: Python, Rust, Scala, C/C++ (CUDA), Go, SQL, JavaScript, Dart
- **Backend & Apps**: FastAPI, Django, React, Flutter, REST APIs, gRPC, PostgreSQL, MongoDB
- **Infrastructure**: Linux, Docker, Kubernetes, Helm, Traefik, AWS, Azure, GCP, Git, CI/CD
- **Data & Systems**: Spark, Kafka, Airflow, Trino, tokio, memory-mapped I/O, benchmarking
- **Spoken Languages**: English (C1), Italian (A2)

Achievements & Test Scores

- **Test Scores**: IELTS 8.0/9.0; GRE Quantitative 165/170
- **Competitive Programming**: ICPC Asia Topi Regionals 2022: 19th place; Google Kickstart (May 2022): 6th place country-wide

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